

# Parallel Programming - Labs 2 to 6

Java implementations, measured results, and raw benchmark files

System: MacBook Air, Apple M3 (8 cores: 4 performance + 4 efficiency), 8 GB unified memory; macOS 15.2; OpenJDK 17.0.18.

This report contains procedure and measurement data only. Analytical question responses are omitted.

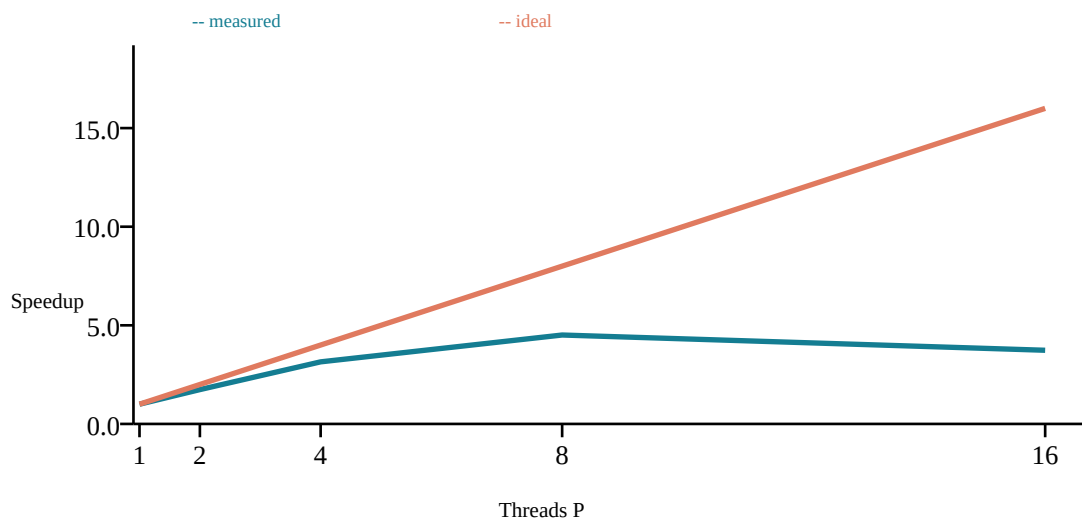
## Lab 2 - Numerical Integration and Reductions

Midpoint integration of pi. Reduction: N=100,000,000, five P values, three measured trials each after a warm-up. Serial baseline: same N. Race and critical-section variants: N=1,000,000; race P=1,2,4,8; critical section P=4. Timer: System.nanoTime().

| Variant   | P  | Trials | Mean ms | Mean abs. error | Mean pi      |
|-----------|----|--------|---------|-----------------|--------------|
| critical  | 4  | 3      | 26.41   | 5.71e-14        | 3.1415926536 |
| race      | 1  | 3      | 2.21    | 2.89e-14        | 3.1415926536 |
| race      | 2  | 3      | 0.51    | 1.55e+00        | 1.5918626529 |
| race      | 4  | 3      | 0.40    | 1.84e+00        | 1.3015926242 |
| race      | 8  | 3      | 0.45    | 1.56e+00        | 1.5812102480 |
| reduction | 1  | 3      | 313.41  | 4.44e-16        | 3.1415926536 |
| reduction | 2  | 3      | 180.08  | 4.44e-16        | 3.1415926536 |
| reduction | 4  | 3      | 99.57   | 4.44e-16        | 3.1415926536 |
| reduction | 8  | 3      | 69.52   | 4.44e-16        | 3.1415926536 |
| reduction | 16 | 3      | 83.91   | 4.44e-16        | 3.1415926536 |
| serial    | 1  | 1      | 61.96   | 6.33e-13        | 3.1415926536 |

Rows in chart order: measured reduction, ideal linear speedup.

### Observed and ideal speedup for parallel reduction



Raw observations: lab2-results.csv. Aggregates: lab2-summary.csv.